

Edgar Cruz-Martinez
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EDUCATION

Georgia Institute of Technology (GT), Atlanta, GA

May 2028

Candidate for Bachelor of Science in Computer Science;

GPA: 3.65 / 4.00

Relevant Coursework: Data Structures and Algorithms, Object and Design, Discrete Math, Computer Organization, and Intro to AI

EXPERIENCE

ACT Lab, *Research Assistant*, GT, Atlanta, GA

January 2025 – Present

- Engineered a Python data-cleaning pipeline using Pandas to automate duplicate detection across 10,000+ autonomous vehicle crash reports, ensuring dataset integrity for downstream analysis.
- Developed an automated taxonomy generator by integrating the GPT API, prompting it to structure unstructured crash narratives into data ready for hierarchical clustering and text mining.
- Generated comprehensive data reports, writing Python scripts to aggregate and visualize key metrics from the dataset.
- Built a custom TraCI controller in Python to interface with SUMO simulations, successfully extracting and visualizing real-time traffic metrics from intersection models.

Walmart, *Apparel Associate*, Gainesville, GA

May 2022 – August 2024

- Oversaw the fitting room's day-to-day operations, making sure that guests felt welcome and that returns and outfit organization were handled effectively.

Brenau University, *Web Development Intern*, Gainesville, GA

August 2023 – May 2024

- Resolved front-end website issues and implemented content updates for high-traffic pages (Home, Events) based on stakeholder requests, utilizing HTML and CMS tools.

PROJECTS

Echo Sign (Computer Vision Game) - Python, Pygame, OpenCV, MediaPipe

- Developed a scale-invariant ASL recognition engine using MediaPipe, normalizing 21 3D hand landmarks against palm length for accurate, depth-agnostic detection.
- Engineered a Euclidean distance algorithm for static sign verification and a multi-frame trajectory buffer to track dynamic, multi-stroke gestures like 'J' and 'Z'.
- Architected a multithreaded backend using Python queues, decoupling computer vision processing from Pygame rendering to filter false positives and maintain a lag-free 60 FPS.

Ripplect (Hackathon Project) - React, TailwindCSS, Leaflet, Python

- Architected the frontend for a crisis simulation dashboard that identifies underfunded nations and predicts how funding changes impact regional displacement across neighboring borders.
- Developed a "Misfit Analysis" metric that revealed a 7x funding disparity between crises; identified "overlooked" regions receiving only \$21 per person compared to \$146 per person in well-funded areas.
- Developed a custom interactive heat map using Leaflet that handles real-time data updates from the backend, allowing users to see immediate changes in crisis severity across borders.
- Implemented a message-syncing system between the dashboard and the map to ensure that simulation results (such as predicted displacement) are rendered correctly without lag or data loss.

Job Board Website - Django, React, SQLite, Agile

- Architected the platform's direct messaging system, enabling real-time communication between recruiters and applicants.
- Developed a Django-based applicant recommendation engine that filters and suggests candidates to recruiters based on specific job-posting criteria.
- Operated within an Agile workflow using Jira, participating in sprint planning, bi-weekly client demos, and retrospectives to iterate on features based on stakeholder feedback.

LEADERSHIP AND COMMUNITY ENGAGEMENT

Every Nation Campus Ministry, *Student President*

September 2024 – Current

- Planned and led small group meetings, planned content for weekly large group meetings, and organized monthly events.

SKILLS

Languages: JavaScript, Python, Java, C, SQL

Libraries & Frameworks: React, Pandas, Django, TailwindCSS, Leaflet, MediaPipe

Tools & Practices: Git, Supabase, Figma, Agile Methodology

HONORS

Honors:

- Pilgrims Tomorrow Fund recipient June 2024
- Hispanic Scholarship Fund recipient July 2024